

Full Stack Development and Deployment of a Web-Based Speech Recognition System

Week 21-23
1 June - 21 June 2026

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Project Overview

- MAGOR is a web-based audio and video transcription system that converts recordings into text
- **Existing Problem:**
 - Although functional, components remain incomplete and require refinement
- **Proposed Solution:**
 - Improve robustness, usability, and AI-driven recording analysis



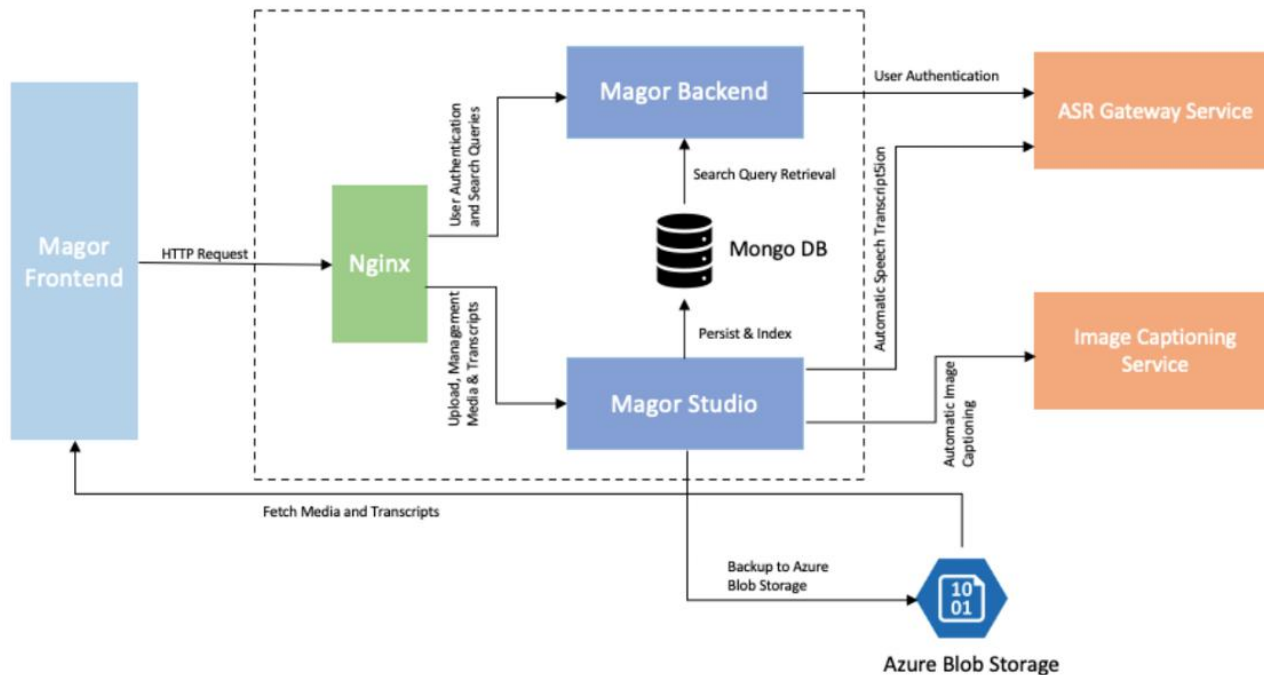
Project Objectives

The primary objectives of this project are:

- Fix system bugs and restore functionality
- Improve usability and user experience
- Enhance maintainability and scalability
- Add AI features to existing pages



Systems Overview



Source: Huang (2021)



Agenda

Work done this week:

1. Admin User Management Page
 - Fixed error when deleting users in bulk
2. Magor Studio
 - Updated transcription job submission to successfully process speaker count through speechlabs gateway
 - Verified tag metrics on the Statistics page
3. Magor Recordings Manager
 - Implemented speaker name change functionality in transcripts
 - Updated UI for the Transcript Editor



Task 1 (Fixed admin user management delete error)

- Error when deleting users in bulk

The screenshot shows a web application interface for managing users. The top navigation bar is blue with a hamburger menu icon, the text "Users", and buttons for "BACK TO MAGOR", a refresh icon, and a user profile icon. A left sidebar contains links for "Users", "recordings", "ASR Requests", and "Statistics". The main content area displays a table of users with columns for "Email", "Name", "Role", and "User status". Each row includes checkboxes for selection, an "EDIT" button, and a "DELETE" button. A status bar at the top of the table indicates "1 item selected" and a "DELETE" button. The table contains 8 rows of user data. The bottom right of the table shows "Rows per page: 10" and "1-8 of 8".

☐	Email	Name	Role	User status		
☐	test@docker.com	test	admin	verified	EDIT	DELETE
☐	testuser1@gmail.com	testuser1	user	unverified	EDIT	DELETE
☐	derick@outlook.com	derick	user	unverified	EDIT	DELETE
☐	newuser@test.com	newUser	user	unverified	EDIT	DELETE
☐	new@test.com	newuser	admin	unverified	EDIT	DELETE
☐	user2@test.com	newuser2	user	unverified	EDIT	DELETE
☒	tee@te.com	tee	user	unverified	EDIT	DELETE
☐	derick@hotmail.com	derick	user	unverified	EDIT	DELETE

Rows per page: 10 1-8 of 8



Task 1 (Fixed admin user management delete error)

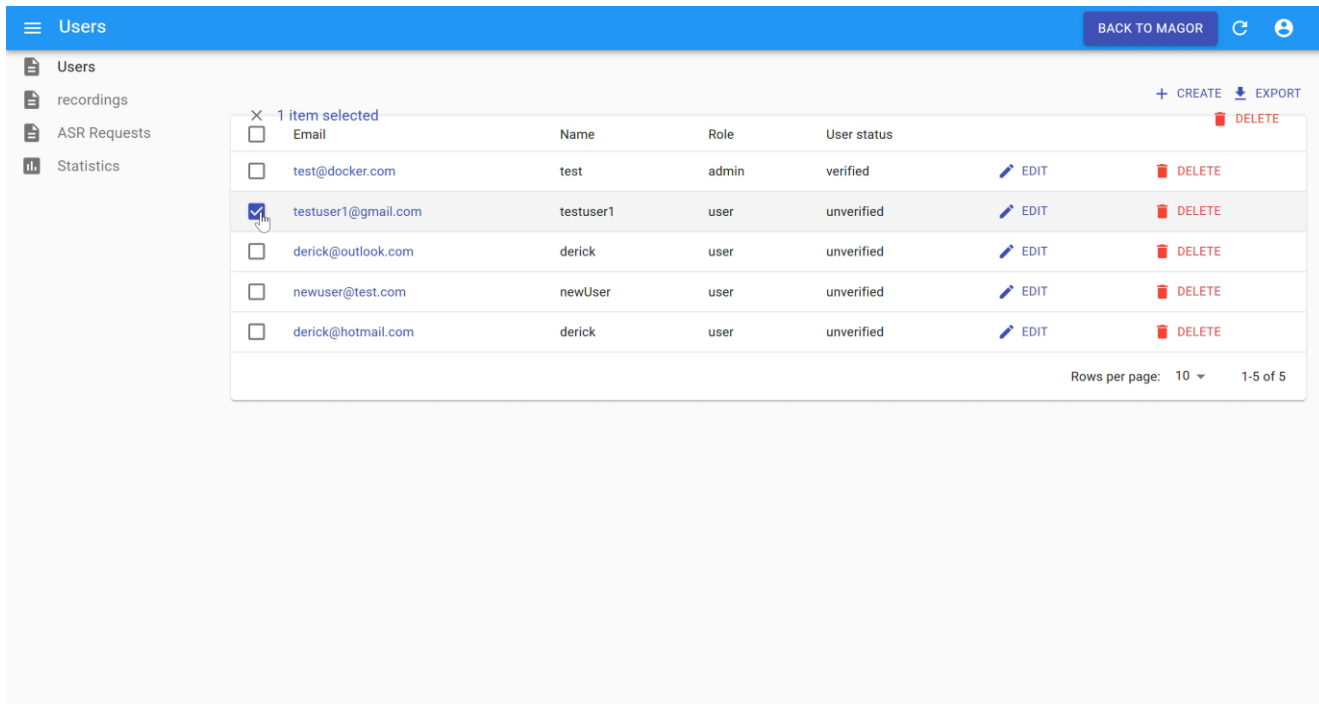
- Error when deleting users in bulk

```
+11 -9  ...  
326  
327     deleteMany: (resource, params) => {  
328         if (resource === 'recordings') {  
329 +             return Promise.all(  
330 +                 params.ids.map((id) => deleteRecordingById(id))  
331 +             ).then(() => ({ data: params.ids })))  
332         }  
333  
334 +         return Promise.all(  
335 +             params.ids.map((id) =>  
336 +                 httpClient(`${apiUrl}/${resource}/${id}`, {  
337 +                     method: 'DELETE',  
338 +                        
339 +                 })  
340 +             ).then(() => ({ data: params.ids })))  
341         },  
342 +         
```



Task 1 (Fixed admin user management delete error)

- Fixed User deletion demo



The screenshot displays a web application interface for managing users. The top navigation bar is blue and contains a menu icon, the text "Users", a "BACK TO MAGOR" button, a refresh icon, and a user profile icon. On the left, a sidebar lists "Users", "recordings", "ASR Requests", and "Statistics". The main content area features a table of users. Above the table, it indicates "1 item selected" and provides "CREATE" and "EXPORT" buttons. The table has columns for "Email", "Name", "Role", and "User status". Each row includes "EDIT" and "DELETE" action buttons. The user "testuser1@gmail.com" is selected, and the "DELETE" button for this user is highlighted in red. At the bottom right of the table, it shows "Rows per page: 10" and "1-5 of 5".

Email	Name	Role	User status
test@docker.com	test	admin	verified
testuser1@gmail.com	testuser1	user	unverified
derick@outlook.com	derick	user	unverified
newuser@test.com	newUser	user	unverified
derick@hotmail.com	derick	user	unverified



Task 2 (Update Transcription Job Request Submission)

- Previous added a number of speakers field:

Auto Transcribe

Current Running Jobs

No jobs running currently.

Send a new transcription job

Language

Abax Default (This model is customised for Singapore English Mandarin Malay)

Audio Type

Close Talk

Audio Track

Single Channel

Number of Speakers

3

Submit

Task 2 (Update Transcription Job Request Submission)

- POST Gateway Request Preview:

```
▼ Request Payload  View source
{
  language: "trilingual",
  audioType: "closetalk",
  audioTrack: "single",
  customData: {num_spk: 2}
}
```

Request Payload

```
X  Headers  Payload  Preview  Response  Initiator  Timing
1  {
  "success": true,
  "transRequest": {
    "status": "pending",
    "_id": "6a25a5c22bcd5f50023ba55b2",
    "mediaId": "6a1c7e26e8906f002330a2a3",
    "audioType": "closetalk",
    "audioTrack": "single",
    "language": "trilingual",
    "customData": {
      "numSpk": -1
    },
    "astId": "6a25a5c23949262333487360",
    "createdAt": "2026-06-07T17:09:22.846Z",
    "updatedAt": "2026-06-07T17:09:22.846Z"
  }
}
```

Mismatched
Response

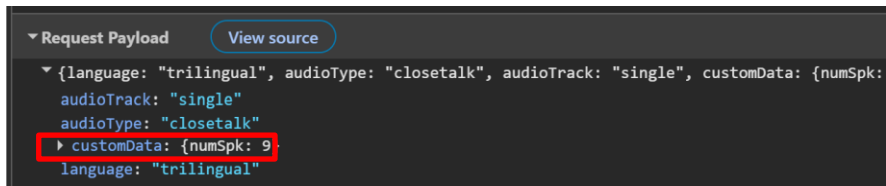
(Gateway not receiving
correctly)



Response

Task 2 (Update Transcription Job Request Submission)

- POST Gateway Request Preview:



```
{language: "trilingual", audioType: "closetalk", audioTrack: "single", customData: {numSpk: 9, language: "trilingual"}}
```

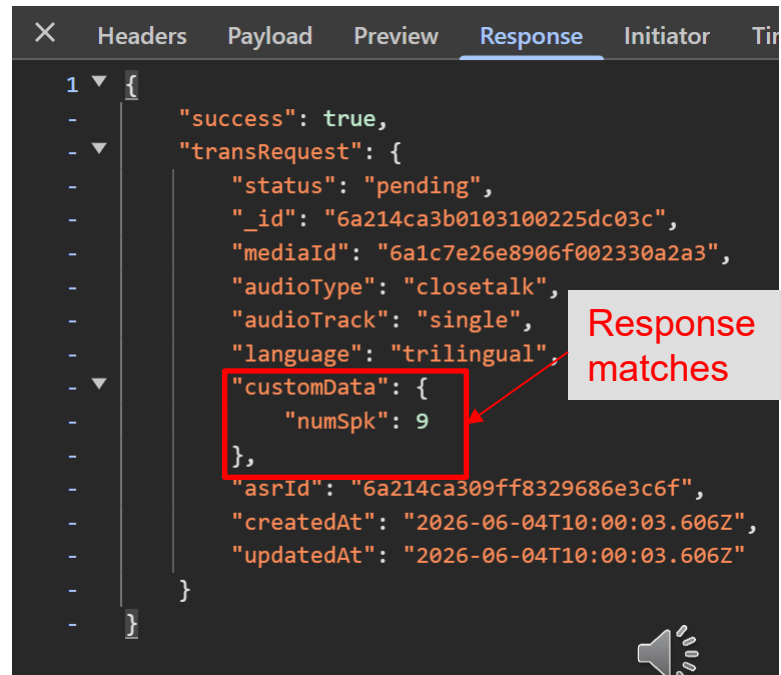
Frontend Request Payload



```
2026-06-04T09:59:52.444Z - info: ASR form body preview:
{
  "lang": "trilingual",
  "audioType": "closetalk",
  "audioTrack": "single",
  "customData[numSpk]": 9,
  "webhook": "http://20.205.178.30/studioapi/asr/webhook",
  "outputFormats": [
    "xml"
  ]
}
```

Use customData[numSpk]

Backend Request Body



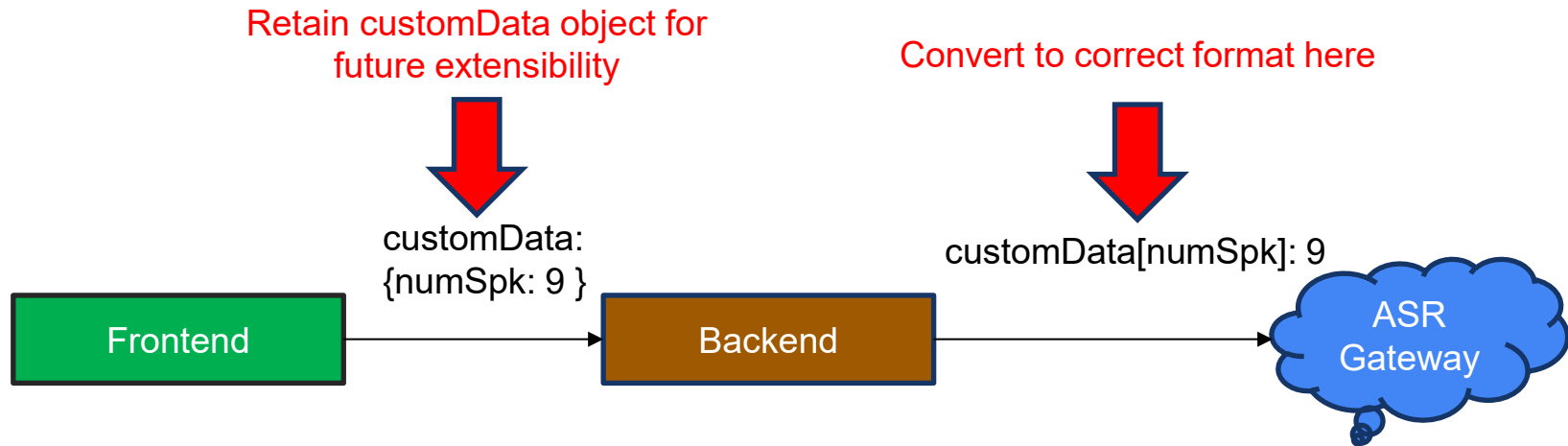
```
{
  "success": true,
  "transRequest": {
    "status": "pending",
    "_id": "6a214ca3b0103100225dc03c",
    "mediaId": "6a1c7e26e8906f002330a2a3",
    "audioType": "closetalk",
    "audioTrack": "single",
    "language": "trilingual",
    "customData": {
      "numSpk": 9
    },
    "asrId": "6a214ca309ff8329686e3c6f",
    "createdAt": "2026-06-04T10:00:03.606Z",
    "updatedAt": "2026-06-04T10:00:03.606Z"
  }
}
```

Response matches

Response Matches

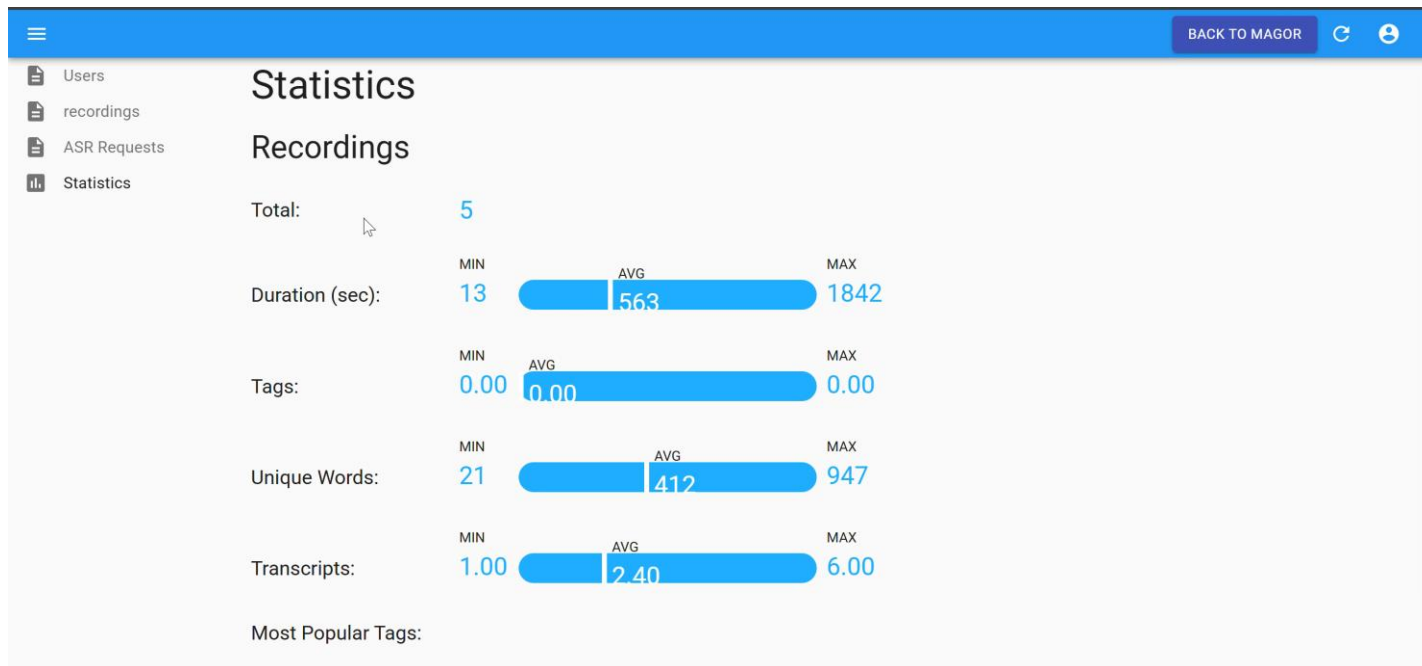
Task 2 (Update Transcription Job Request Submission)

- POST Gateway Request Preview:



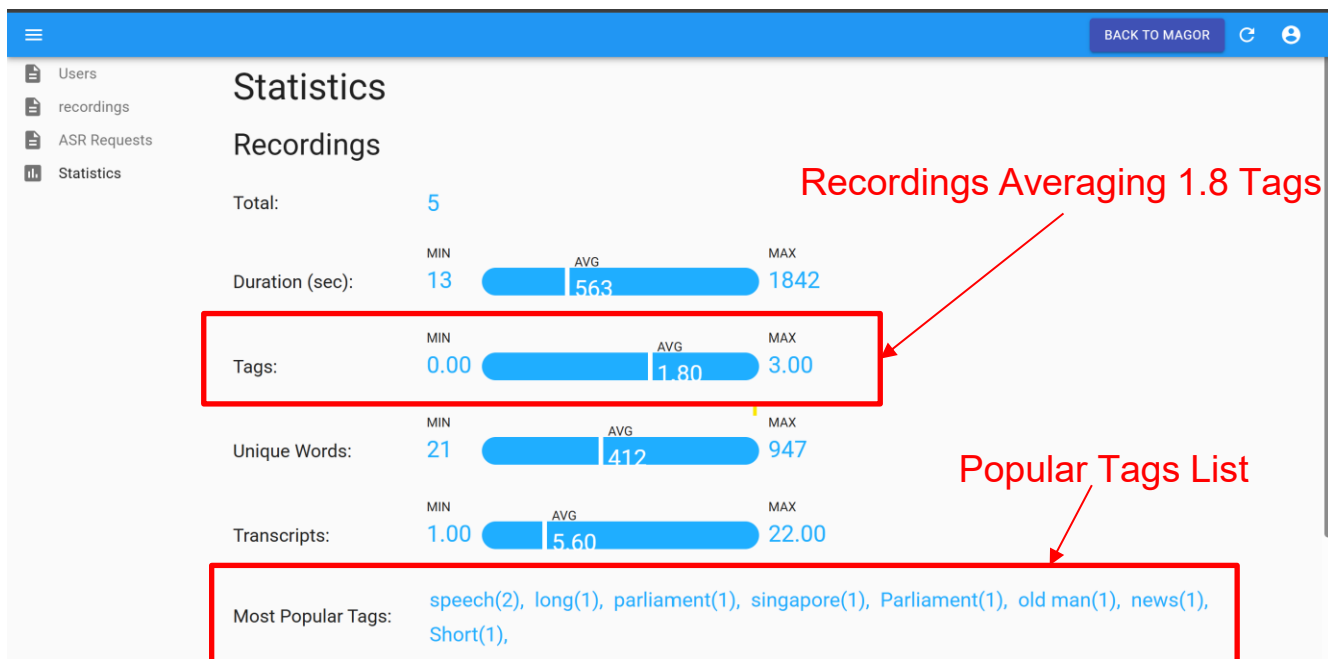
Task 3 (Verify Tags metrics on Statistics Page)

- Previously did not test Tag metrics



Task 3 (Verify Tags metrics on Statistics Page)

- Previously did not test Tag metrics



Task 4 (Recordings Manager Edit Speaker Name Update)

- Previous Transcript Edit Pages:

This recording

Indonesian stocks plunges

Tags: indonesia

Top Words: indonesia concerns msc president shares warning downgrad

Transcript Show options

channel1_S1 oil and business sharp declines for Indonesian shares, the benchmark plunging nearly 9% at one point, triggering a 30 minutes circuit breaker. The Jakarta Composite ending down more than 7% after MSCI warned about the market's investability. DNS provider wants regulators to address concerns that major companies are thinly traded and dominated by a few wealthy owners. It's warning of a possible downgrade if progress is not made by May. Indonesia's rupiah steady today. It has been under sustained pressure for months, tumbling to an all time low against the U.S. dollar.

channel1_S1 currency has been sliding since August 2025 on concerns about fiscal discipline, or is deepening after the resignation of veteran finance Sri Mulliani Indrawati in September the same month. Uh, Bank Indonesia made two surprise rate cuts in line with the government's pro-growth push, raising concerns over its independence. Those concerns deepening in January with the nomination of

Edit Page

This recording

Indonesian stocks plunges

Edit Transcript

Transcript

00:00:00 oil and business sharp declines for Indonesian shares, the benchmark plunging nearly 9% at one point, triggering a 30 minutes circuit breaker. The Jakarta Composite ending down more than 7% after MSCI warned about the market's investability. DNS provider wants regulators to address concerns that major companies are thinly traded and dominated by a few wealthy owners. It's warning of a possible downgrade if progress is not made by May. Indonesia's rupiah steady today. It has been under sustained pressure for months, tumbling to an all time low against the U.S. dollar.

00:00:39 currency has been sliding since August 2025 on concerns about fiscal discipline, or is deepening after the resignation of veteran finance Sri Mulliani Indrawati in September the same month. Uh, Bank Indonesia made two surprise rate cuts in line with the government's pro-growth push, raising concerns over its independence. Those concerns deepening in January with the nomination of President Proposal's nephew for Bank Indonesia's deputy governor with a CNA Channi to explain more about this, uh, stock market route.

Transcript Editor Page

Task 4 (Recordings Manager Edit Speaker Name Update)

- Updated Transcript Edit Pages Demo:

Magor

Search

This recording

test

Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4

Edit

Top Words: indonesia concerns msci bank market point indonesia fiscal president shares warning downgrade progress deepening month

Transcript Show options

channel1_S1| oil and business sharp declines for Indonesian shares, the benchmark plunging nearly 9% at one point, triggering a 30-minute circuit breaker. The Jakarta composite ending down more than 7% after MSCI warned about the market's investability. DNS provider wants regulators to address concerns that major companies are thinly traded and dominated by a few wealthy owners. It's warning of a possible downgrade if progress is not made by May. Indonesia's rupiah steady today, it has been under sustained pressure for months, tumbling to an all-time low against the US dollar.

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Task 4 (Recordings Manager Edit Speaker Name Update)

- Duplicate Speaker Name Protection Demo

Magor

Search

This recording

test

Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4

Edit

Top Words: indonesia concerns msci bank market point indonesia fiscal president shares warning downgrade progress deepening month

Transcript Show options

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Plans for Next Week

- Magor Transcript Studio
 - Implement Set Default button functionality
- Project Maintenance
 - Review and upgrade Node.js versions along with dependencies
 - Research and improve Docker setup and configuration

